

MATH Unit 2 — Differential Calculus Complete Notes

Mathematics (MATH) · Unit 2 · Differential Calculus · 2024

Overview

These notes cover **Differential Calculus**. Engineering and discrete mathematics — linear algebra, calculus, probability, logic and graph theory for computer science. Each section states the core idea, a worked intuition and the points examiners look for.

1. Mean Value Theorems

Rolle's, Lagrange's and Cauchy's mean value theorems relate derivatives to function values.

Why it matters: Mean Value Theorems appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of mean value theorems in one line.
- A small example showing mean value theorems in practice.
- Common mistakes students make and how to avoid them.

2. Taylor and Maclaurin Series

Taylor series expand a function as a power series around a point.

Why it matters: Taylor and Maclaurin Series appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of taylor and maclaurin series in one line.
- A small example showing taylor and maclaurin series in practice.
- Common mistakes students make and how to avoid them.

3. Partial Derivatives

Partial derivatives measure change with respect to one variable while others are fixed.

Why it matters: Partial Derivatives appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of partial derivatives in one line.
- A small example showing partial derivatives in practice.
- Common mistakes students make and how to avoid them.

4. Jacobians

The Jacobian determinant measures how a transformation scales area or volume.

Why it matters: Jacobians appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of jacobians in one line.
- A small example showing jacobians in practice.
- Common mistakes students make and how to avoid them.

5. Maxima and Minima

Critical points of multivariable functions are classified using the second derivative test.

Why it matters: Maxima and Minima appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of maxima and minima in one line.
- A small example showing maxima and minima in practice.
- Common mistakes students make and how to avoid them.

6. Lagrange Multipliers

Lagrange's method finds extrema of a function subject to equality constraints.

Why it matters: Lagrange Multipliers appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of lagrange multipliers in one line.
- A small example showing lagrange multipliers in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Mean Value Theorems · Taylor and Maclaurin Series · Partial Derivatives · Jacobians · Maxima and Minima · Lagrange Multipliers